

Oracle EPM Predictive Cash Forecasting

Statistical Forecasting - Advanced Lab Guide

Product Design Documentation • Implementation Guide • Training Material • Technical Reference • Troubleshooting Handbook

Purpose

This advanced lab guide explains how to design, configure, run, validate and troubleshoot Statistical Forecasting in Oracle EPM Predictive Cash Forecasting using Predictive Planning and Auto Predict.

Design Point	Definition
Forecast method	Statistical Prediction based on historical time-series behavior.
Core Oracle capability	Predictive Planning for interactive forecasting and Auto Predict for scalable batch forecasting.
Typical use case	Cash line items whose future values can be inferred from historical balances or recurring series.
Forecast horizon	Daily, weekly, or monthly depending on application design and data availability.
Output	Base-case prediction, optionally supplemented by best-case and worst-case scenarios.

BISP Trainings

1. Purpose and Learning Outcomes

Learning objective

Students should be able to decide when Statistical Forecasting is appropriate, configure the method, run predictions interactively or in batch, validate confidence and risk indicators, and govern the publication of predicted values into the rolling forecast.

- Understand the role of Statistical Prediction among Oracle PCF forecast methods.
- Explain the difference between Predictive Planning and Auto Predict.
- Identify the history, calendar and data-quality prerequisites for prediction-based forecasting.
- Configure forecast-method assignments for Statistical Prediction.
- Run an interactive prediction from the Predicted tab and interpret the chart, details and risk gauge.
- Use Auto Predict for larger forecasting jobs and write results into target scenarios or versions.
- Validate results against business expectation and perform governed manual overrides where needed.
- Troubleshoot insufficient history, no-result cases and output-quality issues.

Outcome	Advanced student capability
Functional design	Choose statistical forecasting for suitable line items with meaningful historical behavior.
Technical configuration	Enable the feature, set method assignments and prepare historical data correctly.
Operational execution	Run predictions, paste/write back results, and publish forecast versions.
Analytical review	Use risk and variance analysis to separate good predictions from questionable ones.

2. Oracle Product Context

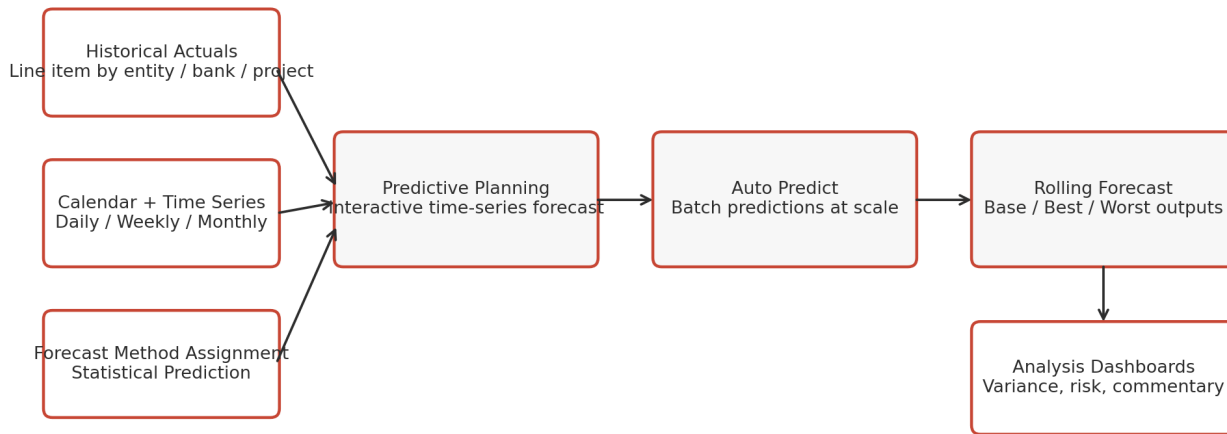
Official Oracle context

Oracle PCF supports Statistical Prediction as a forecast method. It uses Predictive Planning to forecast cash based on historical data using time-series forecasting techniques, and Auto Predict can be used to predict many cells at once when you need scheduled or scalable prediction processing.

Oracle point	Design implication
Statistical Time Series is enabled during application setup	Architects should decide early if prediction-based methods are required.
Predictive Planning selects the best prediction method based on data accuracy	Users focus more on validating business plausibility than manually choosing a formula.
Auto Predict uses the same forecasting/statistical methods as Predictive Planning	Interactive and batch outputs can follow a common modeling philosophy.
Best and Worst cases are available in Auto Predict	Scenario planning can accompany the base-case forecast.
Interactive Predictive Planning may not return results for invoice-only lines without history	Designers should provide alternative reference series or use other methods.

3. Statistical Forecasting Reference Architecture

Statistical Forecasting Reference Architecture



Oracle PCF statistical forecasting uses Predictive Planning and Auto Predict with historical data to produce forecast scenarios.

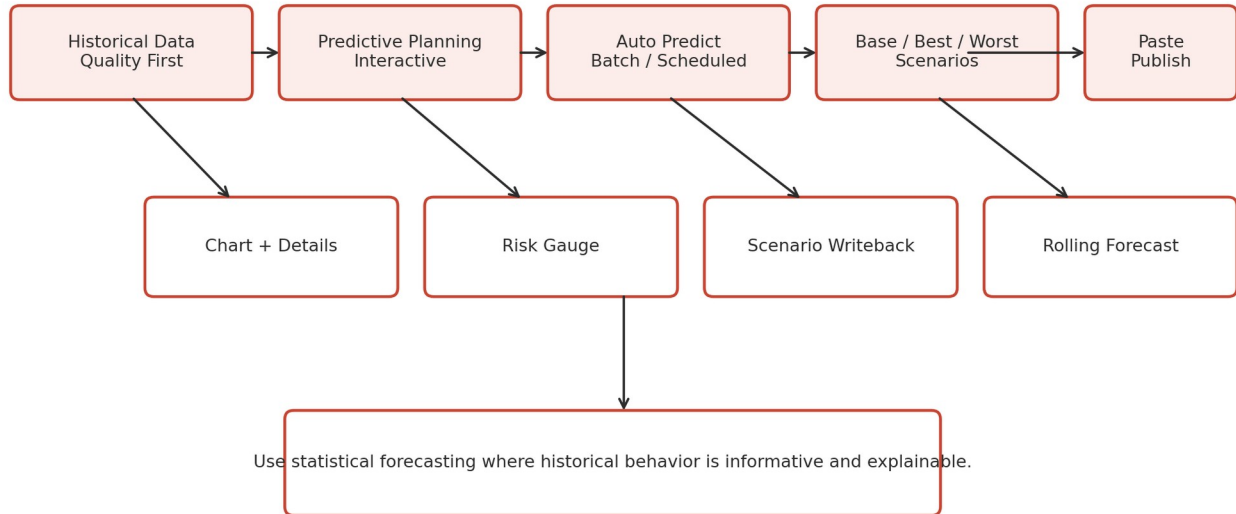
The architecture begins with historical actual data and calendarized time-series storage in the PCF cube. Statistical Prediction is assigned as a forecast method for eligible line items. Predictions are then generated through the interactive Predictive Planning experience or the scheduled Auto Predict engine, and consumed in the Rolling Forecast and dashboards.

Good candidate line items

Line item example	Why statistical forecasting fits
Bank account balances	Historical balances often show repeatable behavior by entity and period.
Utilities / small operating outflows	They may exhibit recurring patterns with modest volatility.
Service fees or bank charges	Useful where cash follows a stable historical series.
Aggregated operating inflows	Appropriate when historical line-level behavior is reasonably smooth and recurring.
Interest income / expense	Can be predicted when balances and rates are historically stable.

4. Statistical Forecasting SmartArt View

Statistical Forecasting SmartArt View

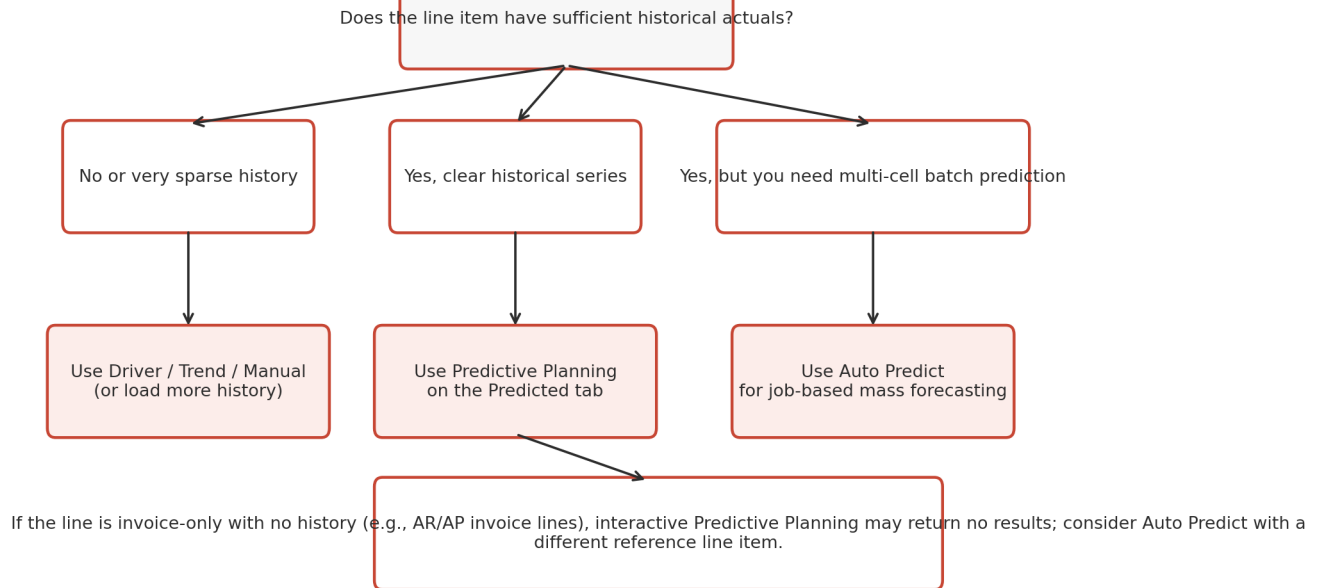


Conceptual summary

Statistical forecasting is a data-driven pattern-recognition approach. High-quality history is the first requirement. The prediction engine then proposes forecast scenarios, which users review before publishing into the rolling forecast.

5. Decision Tree - Choosing the Right Method

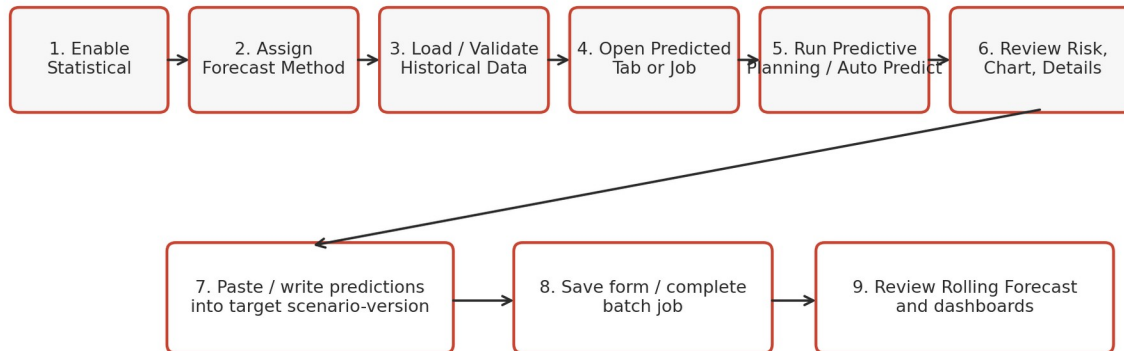
Decision Tree - When to Use Statistical Forecasting



- Use Statistical Forecasting when historical patterns are informative and frequent enough to support prediction.
- Use Predictive Planning when the user wants to interactively inspect one line item or a small set of items.
- Use Auto Predict when you need scheduled processing, higher volume prediction, or writing results in bulk.
- Avoid prediction-based methods for lines with no historical actuals, unless you can reference a more suitable historical series in Auto Predict.
- Fall back to Driver, Trend, Smart Driver or Manual Input when the business logic is better explained by operational drivers than by historical statistical behavior.

6. Process Flow for Statistical Forecasting

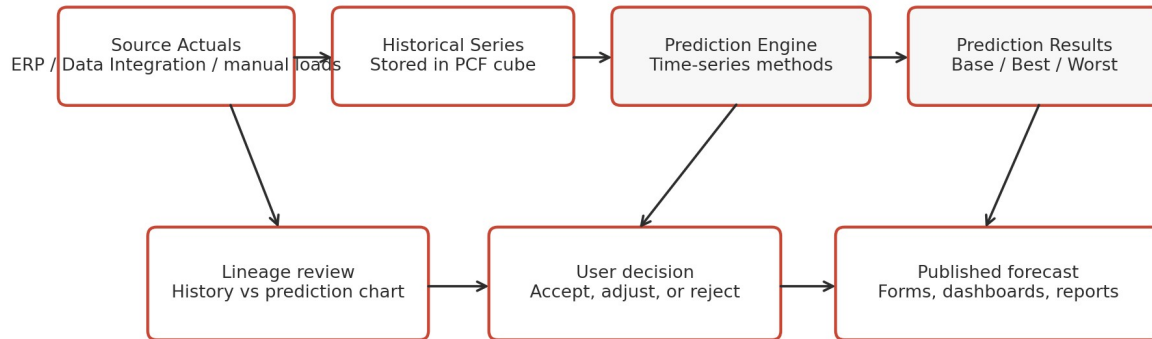
Process Flow - Statistical Forecasting in PCF



1. Enable the Statistical Time Series feature during application setup.
2. Assign Statistical Prediction as the forecast method for eligible line items and entities.
3. Load or validate historical actual data in the correct grain and calendar sequence.
4. Open the Predicted vertical tab for interactive analysis, or configure/run an Auto Predict job for batch processing.
5. Review prediction outputs, including charts, details and scenario/risk information.
6. Paste or write the predicted values into the desired scenario/version and save.
7. Review the resulting cash forecast in Rolling Forecast, dashboards and reports.
8. Capture commentary, approve exceptions, and publish the final version.

7. Data Flow and Prediction Lineage

Data Flow and Prediction Lineage



Control point	Why it matters	Owner
Historical completeness	Missing periods degrade statistical quality or prevent output.	Administrator / data owner
Calendar alignment	Prediction quality depends on correct periodic ordering.	Solution architect
Method assignment	Wrong line-item assignment leads to inappropriate forecasts.	Cash Manager
Write-back target	Users must know where predictions are being stored.	Administrator
Review governance	Predicted values should be distinguishable from overridden values.	Controller / reviewer

8. Oracle-Inspired Configuration Mockup

Oracle-Inspired Configuration Mockup

Home > Application > Enable Features > Predictive Forecasting (Statistical Time Series)

Feature Toggle
Statistical Time Series = ON

Engine
Predictive Planning / Auto Predict

Output Scenario
Forecast

Versions
Base / Best / Worst

Method Assignment
Line item = Statistical Prediction

Prediction Options
Date range / chart settings

Job Controls
Schedule, target cube, results

Save and Validate

Configuration checklist

- ☐ Enable Statistical Time Series for the application.
- ☐ Verify Hybrid Essbase prerequisites where relevant for Auto Predict.
- ☐ Confirm calendars and period structure are valid.
- ☐ Assign Statistical Prediction in Forecast Method Assumptions for suitable line items.
- ☐ Identify output scenario/version for prediction write-back.
- ☐ Decide whether Base only or Base + Best/Worst scenario analysis is required.
- ☐ Security-test access to Predicted forms and any batch jobs.

9. Dashboard and Predicted-Tab Experience

Dashboard Mockup - Predicted and Review Experience



- Interactive users should see the historical series and predicted tail in the same chart.
- The risk gauge should be reviewed, not ignored—it signals uncertainty around the result.
- Details and info boxes should explain the prediction output enough for business review.
- Paste actions should be deliberate and targeted to a clear scenario/version.
- Dashboards should visually distinguish accepted predictions from manual overrides and rejected predictions.

10. Auto Predict Considerations

Oracle consideration	Implementation meaning
Auto Predict works with Cash Forecasting application types when Hybrid Essbase is enabled	Validate environment readiness before promising the feature.
Prediction results are more accurate with more historical data	Data-volume planning is essential in design.
There should be at least twice as much historical data as the number of prediction periods	History depth should be a checkpoint in every implementation workshop.
When there are fewer than six historical points, straight-line fit is used	Short series can produce simplistic results.
With short series, best and worst cases may not be differentiated	Scenario interpretation should acknowledge model limitations.

Key caution

Historical quantity does not guarantee quality. Outliers, broken calendars, major structural changes, and business regime shifts can all weaken statistical relevance.

11. Advanced Lab Scenario

Scenario

A treasury team for Entity IN01 wants to forecast monthly cash balances and selected recurring cash flow line items for the next six months. The team has 18 months of historical actual data and wants to compare interactive Predictive Planning with scheduled Auto Predict.

Lab setting	Value
Entity	IN01
Forecast horizon	Six monthly periods
Historical periods available	18 months
Scenario	Forecast
Versions used	Base, Best, Worst
Target users	Cash Manager, Controller, Administrator

12. Historical Data Used in the Lab

Month	Bank Balance	Utilities	Bank Charges	Service Revenue	Interest Expense
Jan-25	1,220	108	49	940	22
Feb-25	1,180	111	50	955	21
Mar-25	1,260	109	52	968	23
Apr-25	1,305	112	51	980	24
May-25	1,275	114	50	995	24
Jun-25	1,340	113	49	1,005	25
Jul-25	1,360	116	51	1,020	25
Aug-25	1,330	117	50	1,010	26
Sep-25	1,390	115	52	1,028	26
Oct-25	1,420	119	53	1,040	27
Nov-25	1,410	120	51	1,038	28
Dec-25	1,470	122	54	1,065	29

Amounts shown in \$000. Additional months can be assumed in the full data set. The sample is simplified so that advanced learners can focus on the prediction workflow and review mechanics.

13. Lab Step 1 - Assign Forecast Methods

Line Item	Forecast Method	Reason
Bank Balance	Statistical Prediction	Historical series is strong and suitable for pattern-based prediction.
Utilities	Statistical Prediction	Stable recurring pattern with mild seasonality.
Bank Charges	Statistical Prediction	Consistent low-volatility series.
Service Revenue Cash Inflow	Statistical Prediction	Good historical continuity with upward pattern.
Interest Expense	Statistical Prediction	Small but recurring predictable series.

14. Lab Step 2 - Interactive Predictive Planning Workflow

1. From Home, open Daily Cash Forecast or Periodic Cash Forecast based on the application design.
2. Click the Predicted vertical tab.
3. Select the appropriate POV members such as Entity, Scenario, Version, Currency and Line Item.
4. Select a line item, then choose Predictive Planning from the Actions menu.
5. Review the resulting chart below the form, which displays history and forecast.
6. Use tooltips, info boxes, risk gauge and details for deeper review.
7. Customize chart or prediction options if needed.
8. Paste predictions to the selected scenario/version and save the form.

Interactive benefit

Interactive Predictive Planning is ideal when the user wants explainability at line-item level—seeing the chart, details and risk before accepting the predicted values.

15. Lab Step 3 - Auto Predict Batch Workflow

1. Define an Auto Predict job to process the required slice of data.
2. Specify source history and target scenario/version for results.
3. Select the output requirement, such as Base only or Base + Best + Worst.
4. Run the job on demand or schedule it for the regular forecasting cycle.
5. Review job completion logs and any warnings.
6. Confirm that predicted values were written to the intended version.
7. Review the results in Rolling Forecast and dashboards.
8. Escalate exceptions where results appear implausible or history was weak.

Auto Predict design point	Recommendation
Frequency	Align with forecasting cadence (daily, weekly, monthly).
Slice design	Start small during UAT; scale gradually in production.
Target write-back	Keep test and production versions separate.
Exception review	Flag lines with low history depth or sudden pattern changes.

16. Step-by-Step Example Forecast

Example line item: Bank Balance

The line shows a broadly rising pattern with some month-to-month variation. Statistical Forecasting uses the historical time-series to derive the next periods. The user is not required to choose the formula manually; instead, the user validates the output and decides whether to accept it.

Future Month	Base Prediction	Best Case	Worst Case	Review Note
Jan-26	1,490	1,525	1,455	Base case follows trend continuation.
Feb-26	1,505	1,545	1,465	Slight uplift expected.
Mar-26	1,530	1,575	1,485	Quarter-end build reflected.
Apr-26	1,548	1,595	1,500	Consistent with prior seasonal lift.
May-26	1,538	1,585	1,490	Mild normalization.
Jun-26	1,565	1,615	1,510	Ends above prior year June.

Interpretation: the base case becomes the working forecast candidate, while the best and worst cases support risk-based discussion rather than immediate posting as the operational forecast.

17. Prediction Review and Validation

Validation question	Reviewer action
Does the shape of the forecast resemble an understandable business story?	Compare with historical chart and current business context.
Are best and worst scenarios too narrow or too wide?	Judge forecast uncertainty and whether data quality is adequate.
Did the model ignore a known structural break?	Consider manual override or alternate method.
Is the prediction materially different from the prior forecast?	Add commentary and explain operational changes.
Was the wrong line item predicted due to no history?	Reassign method or use Auto Predict reference-series logic.

Golden rule

Statistical output is a decision aid, not a replacement for business review. The best implementations treat predictions as explainable recommendations subject to governance.

18. Forecast Accuracy and Risk Interpretation

Indicator	Meaning	How to use it
Historical fit quality	How well the method explains the past pattern	Use as one input, not the only one.
Risk gauge	Relative certainty or uncertainty of the selected forecast	Escalate high-risk items for review.
Scenario spread	Distance between best and worst cases	Wide spreads imply caution in planning decisions.
Variance to prior forecast	Difference from the last approved forecast	Use to detect model drift or real business change.
Variance to actual after close	Ex-post accuracy feedback	Use for continuous improvement of method assignments.

19. Limitations and Design Considerations

- Prediction quality deteriorates when history is sparse, noisy or structurally inconsistent.
- Statistical Forecasting works best for historically patterned line items; it is weaker for unique events, restructurings or one-off financing flows.
- Interactive Predictive Planning may not produce results for Receivable Invoices, Overdue Invoices, Payable Invoices and Payable Overdue Invoices because those lines do not provide suitable historical data in that interactive context.
- Auto Predict may still be useful in cases where you can reference a more appropriate historical line item or process a larger set systematically.
- Users must separate model quality issues from business-change issues—sometimes the model is fine, and the business really has changed.

20. Troubleshooting Handbook

Log / Error Examples

ERROR STF-101 Statistical Time Series feature is not enabled for this application.

WARN STF-118 History length is shorter than recommended; results may use straight line fit.

ERROR STF-204 No historical actual data found for selected line item in interactive Predictive Planning.

INFO STF-001 Auto Predict job completed. Base, Best and Worst results written to target version.

Symptom	Likely cause	Resolution
No results in Predictive Planning	No historical actuals for the selected line item	Choose a different eligible line item or use another method.
Very linear-looking forecast	Too few history points or no strong pattern	Increase history depth and review data quality.
Best/Worst not meaningful	Short or weak series	Use base case cautiously and document limitations.
Prediction job not available	Feature not enabled or environment not ready	Enable Statistical Time Series and validate prerequisites.
Results written to wrong version	Incorrect job or paste target	Strengthen write-back governance and testing.
Forecast rejected by business users	Output is mathematically valid but operationally implausible	Capture commentary, override if justified, and review method assignment.

21. Product Design Documentation Notes

Design area	Recommendation
Navigation	Expose Predicted as a clear vertical tab with nearby Rolling Forecast access.
User experience	Show chart, risk, details and paste action together to support fast review.
Auditability	Persist or report whether forecast values originated from prediction or override.
Scenario architecture	Separate Base operational forecast from Best/Worst scenario analysis.
Security	Restrict who can run jobs, paste results, and publish versions.

22. Roles and Responsibilities

Role	Responsibilities
Administrator	Enable features, maintain artifacts, support jobs, manage security and data readiness.
Cash Manager	Run interactive predictions, review results, paste values and provide commentary.
Controller	Review exceptions, approve overrides and validate governance.
Solution Architect	Define method-selection standards and production design principles.
Support Team	Monitor logs, batch failures, history gaps and user issues.

23. Production Runbook

1. Load or validate actual historical data for the active forecasting cycle.
2. Confirm the line items assigned to Statistical Prediction remain appropriate.
3. Run interactive predictions for key review lines or launch the scheduled Auto Predict job.
4. Check logs and investigate warnings about history depth or missing data.
5. Review base, best and worst outputs in forms and dashboards.
6. Accept, reject or override predictions according to policy.
7. Publish the approved forecast version and archive commentary/evidence.
8. After actuals close, compare results to improve next cycle accuracy.

24. Implementation Checklist

- ☐ Enable Statistical Time Series during application setup.
- ☐ Confirm calendar and period structure.
- ☐ Validate Hybrid Essbase / environment prerequisites for Auto Predict where applicable.
- ☐ Define line-item eligibility criteria for statistical forecasting.
- ☐ Assign Statistical Prediction in the Forecast Method dimension/form.
- ☐ Prepare sufficient historical actual data.
- ☐ Test interactive Predictive Planning on representative line items.
- ☐ Test Auto Predict job design and write-back targets.
- ☐ Design dashboards, reports and exception views.
- ☐ Train users on chart interpretation, risk review and governance of overrides.

25. Advanced Student Challenge

Challenge case

Entity US01 has 24 months of history for Bank Balance and Utilities, 5 months of history for a new service-fee line, and no direct historical actuals for an invoice-only cash line. The student must decide which lines should use Statistical Forecasting and which should use alternate methods.

- Recommend the correct method for each line item and justify it.
- Explain when Predictive Planning is enough and when Auto Predict is preferable.
- Show how you would handle the five-month service-fee series, acknowledging the straight-line-fit limitation.
- Describe what dashboard, control and log views you would provide to the Cash Manager and Controller.

26. Summary and Completion Criteria

- The learner understands Statistical Prediction as a forecast method in Oracle PCF.
- The learner can differentiate Predictive Planning and Auto Predict use cases.
- The learner can configure, run, validate and publish statistical forecasts in a governed process.
- The learner can identify lines where prediction-based methods are unsuitable or risky.
- The document serves as product design documentation, implementation guide, training material, technical reference and troubleshooting handbook.

Completion state

A successful implementation of Statistical Forecasting improves forecast speed and consistency while preserving business review, auditability and explanation of results.